In [2]:
```
!pip install seaborn pandas matplotlib
```

```
Collecting seaborn
  Downloading seaborn-0.13.2-py3-none-any.whl.metadata (5.4 kB)
Requirement already satisfied: pandas in c:\users\n tanusri\appdata\local\program
s\python\python313\lib\site-packages (3.0.3)
Requirement already satisfied: matplotlib in c:\users\n tanusri\appdata\local\pro
grams\python\python313\lib\site-packages (3.11.0)
Requirement already satisfied: numpy!=1.24.0,>=1.20 in c:\users\n tanusri\appdata
\local\programs\python\python313\lib\site-packages (from seaborn) (2.5.1)
Requirement already satisfied: python-dateutil>=2.8.2 in c:\users\n tanusri\appda
ta\local\programs\python\python313\lib\site-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: tzdata in c:\users\n tanusri\appdata\local\program
s\python\python313\lib\site-packages (from pandas) (2026.3)
Requirement already satisfied: contourpy>=1.0.1 in c:\users\n tanusri\appdata\loc
al\programs\python\python313\lib\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in c:\users\n tanusri\appdata\local\p
rograms\python\python313\lib\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.22.0 in c:\users\n tanusri\appdata\lo
cal\programs\python\python313\lib\site-packages (from matplotlib) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in c:\users\n tanusri\appdata\lo
cal\programs\python\python313\lib\site-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in c:\users\n tanusri\appdata\loca
l\programs\python\python313\lib\site-packages (from matplotlib) (26.2)
Requirement already satisfied: pillow>=9 in c:\users\n tanusri\appdata\local\prog
rams\python\python313\lib\site-packages (from matplotlib) (12.3.0)
Requirement already satisfied: pyparsing>=3 in c:\users\n tanusri\appdata\local\p
rograms\python\python313\lib\site-packages (from matplotlib) (3.3.2)
Requirement already satisfied: six>=1.5 in c:\users\n tanusri\appdata\local\progr
ams\python\python313\lib\site-packages (from python-dateutil>=2.8.2->pandas) (1.1
7.0)
Downloading seaborn-0.13.2-py3-none-any.whl (294 kB)
Installing collected packages: seaborn
Successfully installed seaborn-0.13.2
```

```
[notice] A new release of pip is available: 25.2 -> 26.2.1
[notice] To update, run: python.exe -m pip install --upgrade pip
```

In [3]:
```
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt

df = sns.load_dataset('titanic')
df.head()
```

Out[3]:

	survived	pclass	sex	age	sibsp	parch	fare	embarked	class	who	adul
0	0	3	male	22.0	1	0	7.2500	S	Third	man	
1	1	1	female	38.0	1	0	71.2833	C	First	woman	
2	1	3	female	26.0	0	0	7.9250	S	Third	woman	
3	1	1	female	35.0	1	0	53.1000	S	First	woman	
4	0	3	male	35.0	0	0	8.0500	S	Third	man	

In [4]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 891 entries, 0 to 890
Data columns (total 15 columns):
 #   Column       Non-Null Count  Dtype
---  ------       --------------  -----
 0   survived     891 non-null    int64
 1   pclass       891 non-null    int64
 2   sex          891 non-null    str
 3   age          714 non-null    float64
 4   sibsp        891 non-null    int64
 5   parch        891 non-null    int64
 6   fare         891 non-null    float64
 7   embarked     889 non-null    str
 8   class        891 non-null    category
 9   who          891 non-null    str
 10  adult_male   891 non-null    bool
 11  deck         203 non-null    category
 12  embark_town  889 non-null    str
 13  alive        891 non-null    str
 14  alone        891 non-null    bool
dtypes: bool(2), category(2), float64(2), int64(4), str(5)
memory usage: 100.4 KB
```

In [5]:
```
df.describe()
```

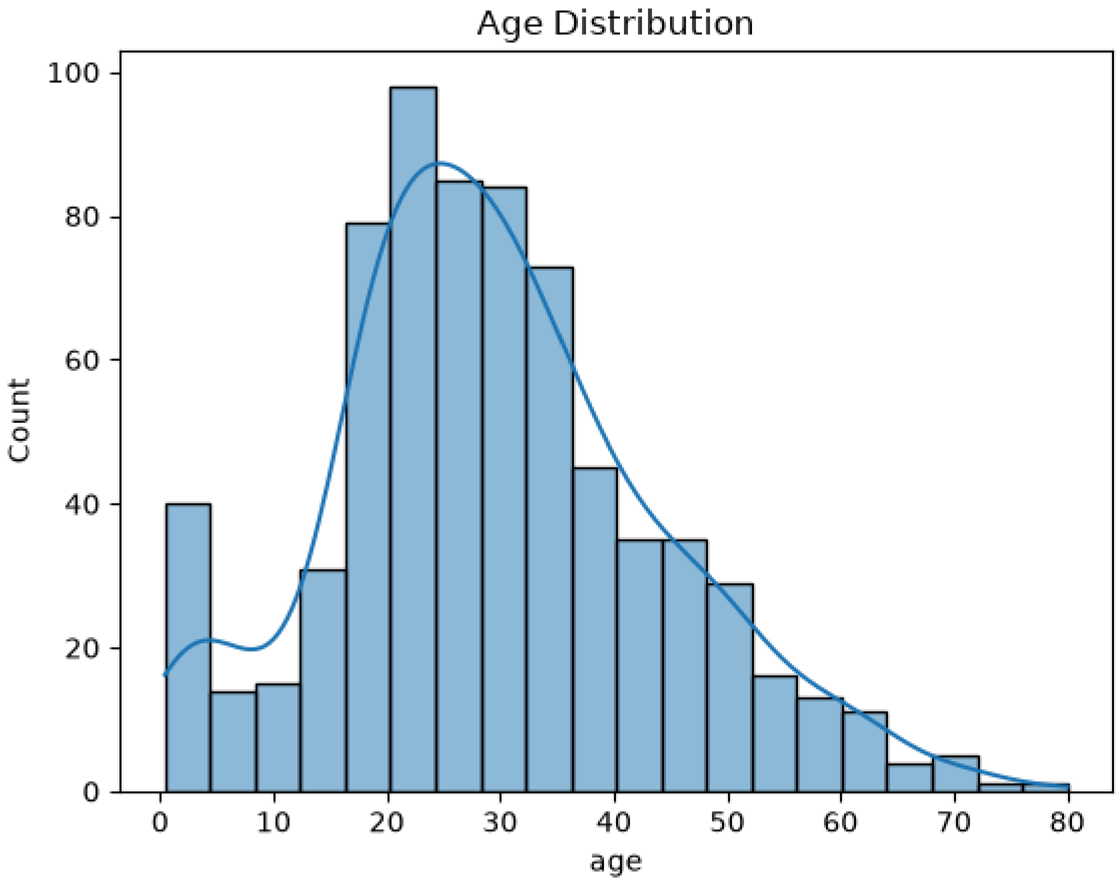
Out[5]:

	survived	pclass	age	sibsp	parch	fare
count	891.000000	891.000000	714.000000	891.000000	891.000000	891.000000
mean	0.383838	2.308642	29.699118	0.523008	0.381594	32.204208
std	0.486592	0.836071	14.526497	1.102743	0.806057	49.693429
min	0.000000	1.000000	0.420000	0.000000	0.000000	0.000000
25%	0.000000	2.000000	20.125000	0.000000	0.000000	7.910400
50%	0.000000	3.000000	28.000000	0.000000	0.000000	14.454200
75%	1.000000	3.000000	38.000000	1.000000	0.000000	31.000000
max	1.000000	3.000000	80.000000	8.000000	6.000000	512.329200

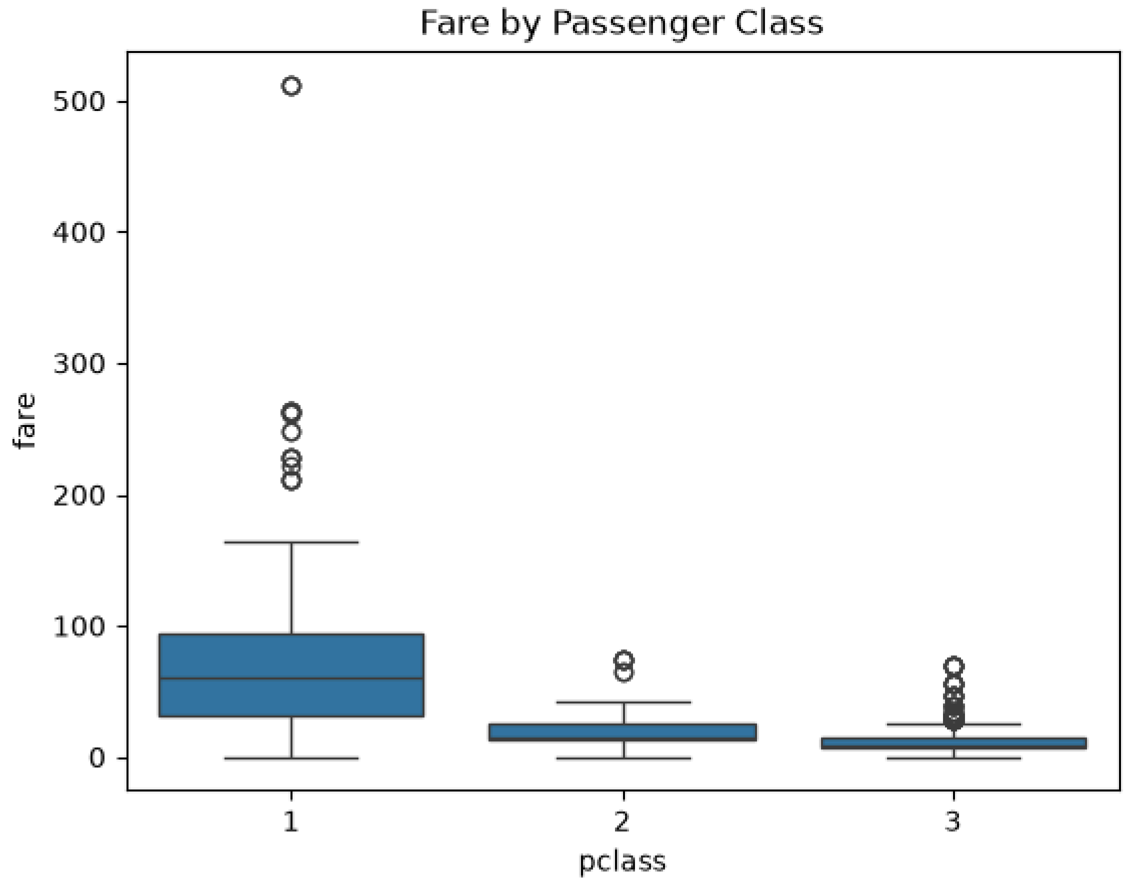
In [7]:
```
print(df['survived'].value_counts())
print(df['sex'].value_counts())
print(df['pclass'].value_counts())
```

```
survived
0    549
1    342
Name: count, dtype: int64
sex
male      577
female    314
Name: count, dtype: int64
pclass
3    491
1    216
2    184
Name: count, dtype: int64
```

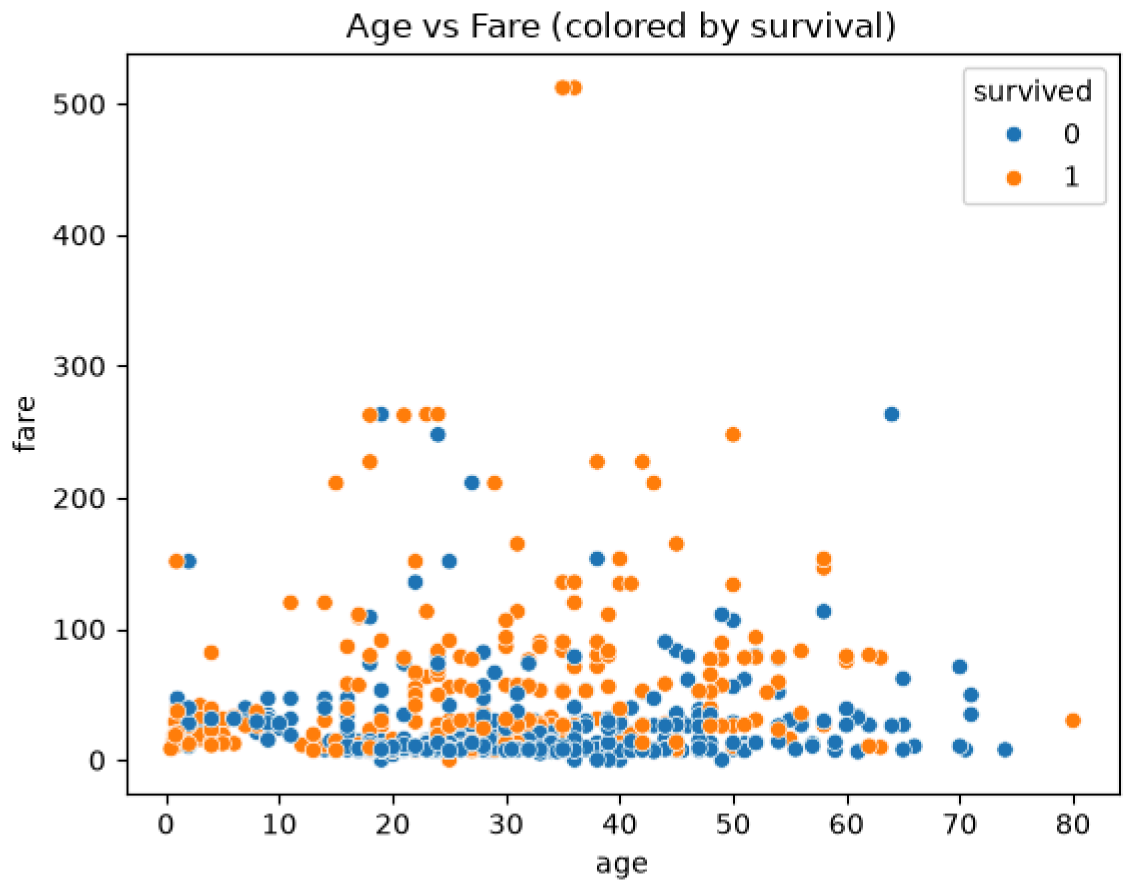
In [8]:
```
sns.histplot(df['age'].dropna(), kde=True)
plt.title('Age Distribution')
plt.show()
```



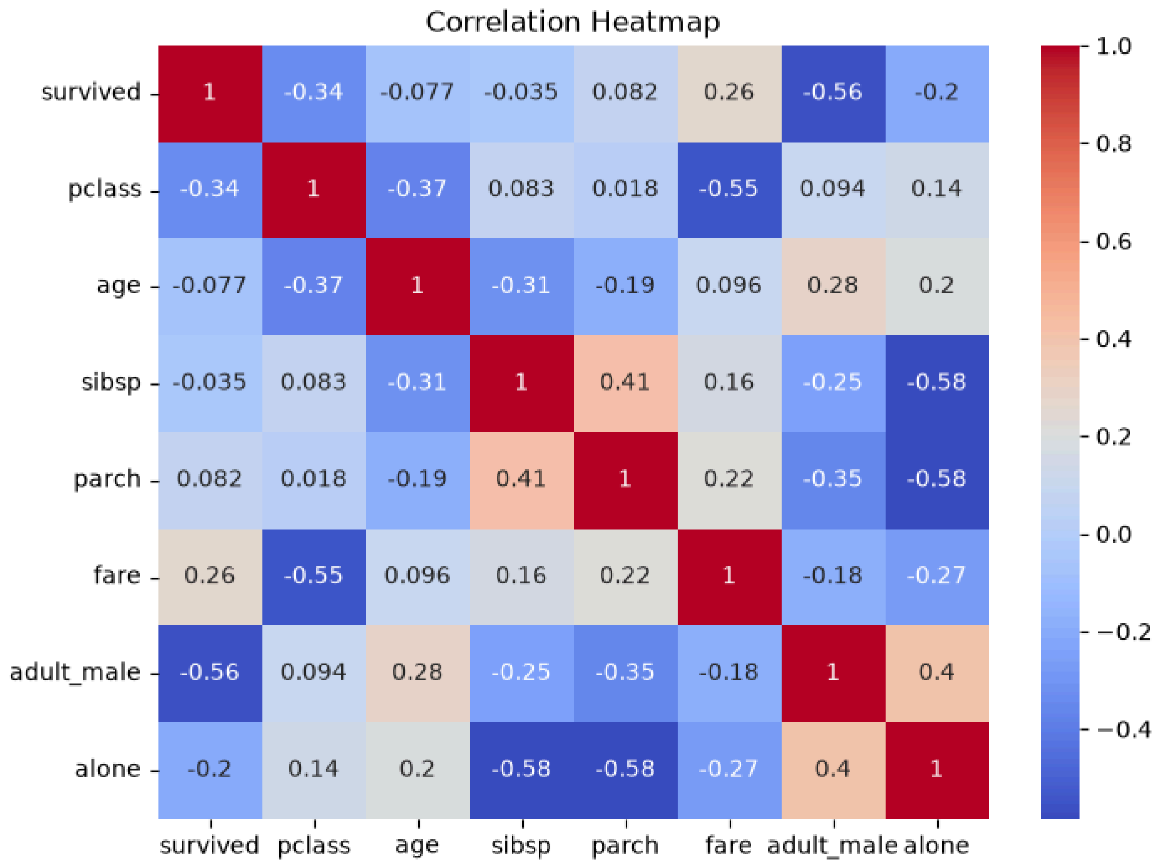
In [10]:
```
sns.boxplot(x='pclass', y='fare', data=df)
plt.title('Fare by Passenger Class')
plt.show()
```



In [11]:
```
sns.scatterplot(x='age', y='fare', hue='survived', data=df)
plt.title('Age vs Fare (colored by survival)')
plt.show()
```



In [12]:
```
plt.figure(figsize=(8,6))
sns.heatmap(df.corr(numeric_only=True), annot=True, cmap='coolwarm')
plt.title('Correlation Heatmap')
plt.show()
```

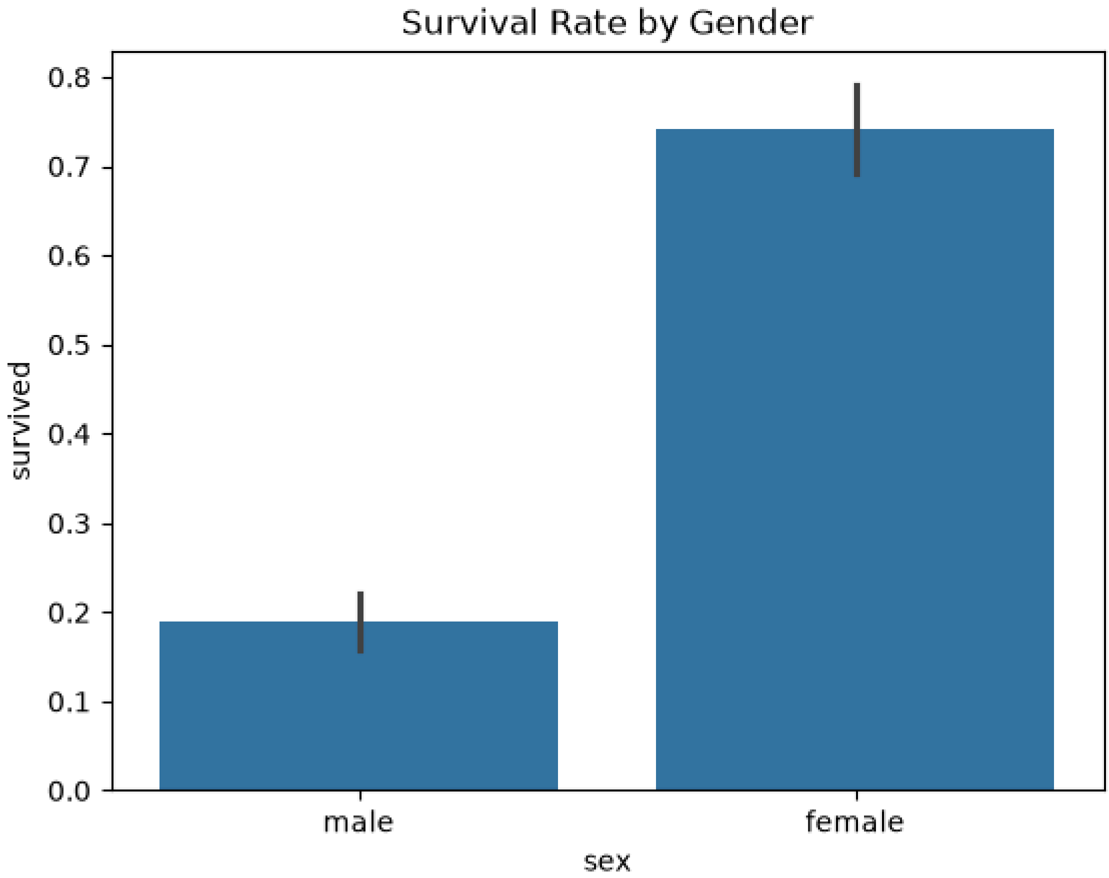


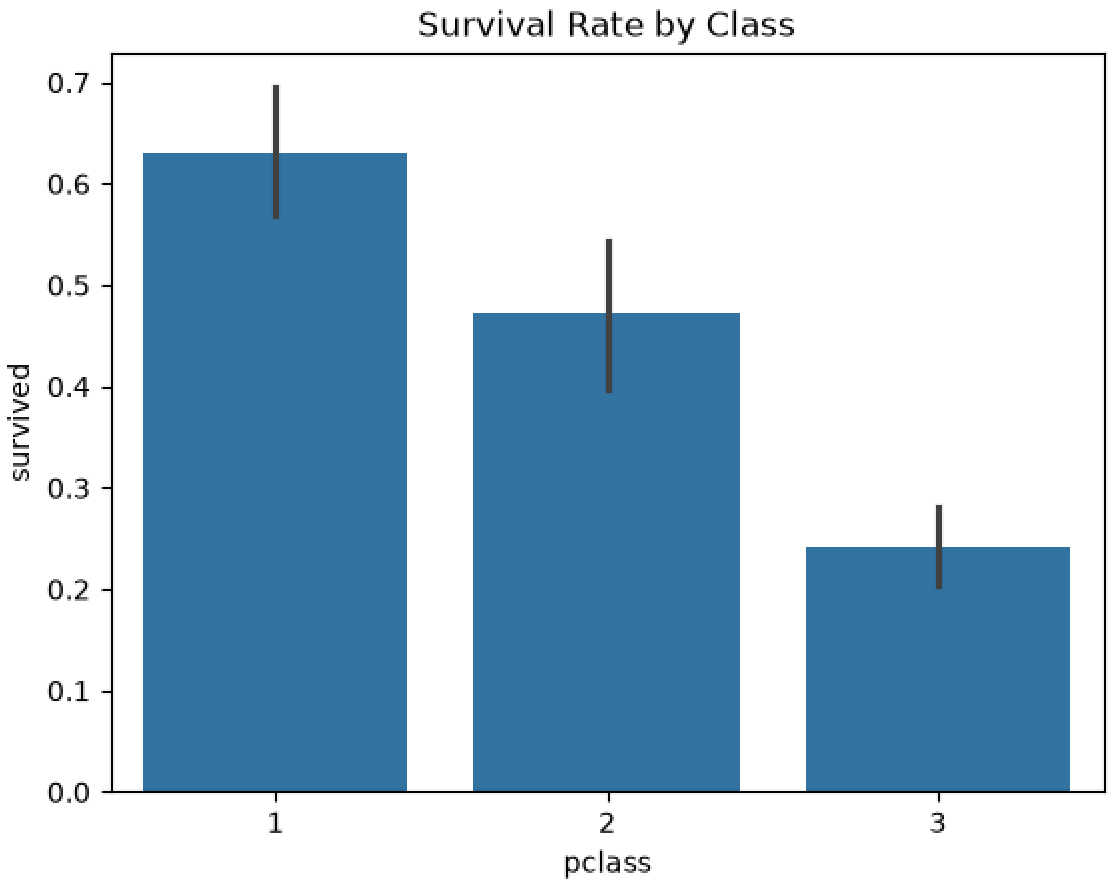
In [14]:
```
print(df.groupby('sex')['survived'].mean())
print(df.groupby('pclass')['survived'].mean())
```

```
sex
female    0.742038
male      0.188908
Name: survived, dtype: float64
pclass
1    0.629630
2    0.472826
3    0.242363
Name: survived, dtype: float64
```

In [15]:
```
sns.barplot(x='sex', y='survived', data=df)
plt.title('Survival Rate by Gender')
plt.show()

sns.barplot(x='pclass', y='survived', data=df)
plt.title('Survival Rate by Class')
plt.show()
```





Summary of Findings

- Survival rate by gender showed the biggest gap: **74.2% of females survived vs only 18.9% of males** — strong evidence of "women and children first."
- Survival also dropped steadily by class: **1st class 63%, 2nd class 47%, 3rd class 24%** — wealthier/higher-class passengers had much better survival odds.
- Overall, about 38% of all passengers survived.
- Most passengers traveled 3rd class (491 of 891), and there were nearly twice as many male as female passengers.
- Fare varied greatly by class — 1st class fares were far higher and more spread out (up to $512), while 3rd class fares were low and tightly clustered.
- Passengers who survived tended to have paid higher fares, reinforcing the class/wealth-survival link.
- Age had 177 missing values and deck had 688 missing — a data quality note.
- Age was roughly bell-shaped, peaking around 20-30 years old.

In [ ]:

```
```